

Modular organization of balance control following perturbations during walking

Anderson Souza Castelo Oliveira,^{1,2} Leonardo Gizzi,^{3,4} Uwe Gustav Kersting,¹ and Dario Farina⁴

¹Center for Sensory-Motor Interaction, Department of Health Science and Technology, Aalborg University, Aalborg, Denmark; ²The CAPES Foundation, Brazilian Education Ministry, Brasilia, Brazil; ³Dipartimento di Scienze del Movimento Umano e della Salute, Università degli studi di Roma "Foro Italico," Rome, Italy; and ⁴Department of Neurorehabilitation Engineering, Bernstein Focus Neurotechnology Göttingen, Bernstein Center for Computational Neuroscience, University Medical Center Göttingen, Georg-August University, Göttingen, Germany

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Oliveira AS, Gizzi L, Kersting UG, Farina D. Modular organization of balance control following perturbations during walking. *J Neurophysiol* 108: 1895–1906, 2012. First published July 5, 2012; doi:10.1152/jn.00217.2012.—Balance recovery during walking requires complex sensory-motor integration. Mechanisms to avoid falls are active concomitantly with human locomotion motor patterns. It has been suggested that gait can be described by a set of motor modules (synergies), but little is known on the modularity of gait during recovery of balance due to unexpected slips. Our hypothesis was that muscular activation during reactive recovery of balance during gait has a modular organization. The aim of the study was to verify this hypothesis when perturbations were delivered in different directions. Eight healthy men walked on a 7-m walkway, which had a moveable force platform embedded in the middle. Subjects experienced unperturbed walking as well as perturbations delivered in the sagittal (forward and backward) and frontal (leftward and rightward) planes. Bilateral full-body kinematics and surface electromyography (EMG) from lower limbs, trunk, and neck were recorded during walking. Synergies and activation signals were extracted from surface EMG signals. Four modules were sufficient to explain the unperturbed gait and the gait perturbed in any of the perturbation directions. Moreover, three of four modules extracted from the unperturbed gait were the same for gait perturbed forward, leftward, and rightward (similarity in synergies = 0.94 ± 0.03). On the other hand, the activation signals were different between unperturbed and perturbed gait (average correlation coefficient = 0.55 ± 0.16). These strategies to recover balance were robust across subjects. In conclusion, changes in lower limb and trunk kinematics provoked by perturbations were reflected in minimal adjustments in the muscular modular organization of walking, with three of four modules preserved from normal walking. Conversely, the activation signals were all substantially influenced by the perturbations, being the result of integration of afferent information and supraspinal control.

balance; EMG; gait; synergies

IT HAS BEEN SUGGESTED that the central nervous system (CNS) may simplify the control of complex movements by activating a limited number of motor modules, also called muscle synergies (Ivanenko et al. 2004, 2005; Muceli et al. 2010; Ting and McKay 2007; Torres-Oviedo and Ting 2007). Motor modules have been identified in the control of natural motor behaviors, such as swimming and kicking in frogs (d'Avella et al. 2003)

as well as during human locomotion (Cappellini et al. 2006; Gizzi et al. 2011; Ivanenko et al. 2004, 2005, 2006; Monaco et al. 2010). In these studies, human walking could be described by a set of four to six modules associated with locomotion at different speeds (Ivanenko et al. 2004, 2005). Interestingly, the inclusion of a voluntary action during walking, such as kicking, could be described by adding one motor module, which was interpreted as a superposition of a task-specific module with the locomotion modules (Ivanenko et al. 2005).

Walking is substantially altered when balance is compromised. For example, Cappellini et al. (2010) observed increased overall muscular activity during walking on a slippery surface. Indeed, maintenance of posture and balance for humans involves complex sensory-motor integration (Torres-Oviedo and Ting 2007, 2010). Similar to walking, postural adjustments during quiet standing have been described by few motor modules, which are consistent across different directions of perturbation (see Torres-Oviedo and Ting 2007, 2010 for details). However, little is known on the motor control strategies associated with maintenance of balance following an unexpected perturbation during walking.

Perturbations to walking such as those due to stumbling or to the effect of tilting support surfaces require muscular reactions in different phases of the gait cycle (Pijnappels et al. 2005; Schillings et al. 2000) and might also require more than five steps until a complete recovery (Oddsson et al. 2004). The interlimb coordination in this case has to sequentially provide time and clearance for the positioning of the recovery limb (Pijnappels et al. 2004), which might not be essential in other perturbation conditions such as slipping. Slippery surfaces lead individuals to assume a more cautious gait pattern, which has different body kinematics and muscular activity throughout the gait cycle (Cappellini et al. 2010; Chambers and Cham 2007). Usually natural slips forward begin 50 ms after a heel strike at high velocity, which induces sliding (Chambers and Cham 2007; Redfern et al. 2001). These perturbations to balance are likely to result in falls if the slip distance exceeds 10 cm (Cham and Redfern 2002; Redfern et al. 2001) or if the sliding velocity is greater than ~ 50 cm/s (Redfern et al. 2001). These parameters (slipping onset, distance, and velocity) observed during natural slips can be controlled by using moving surfaces to induce forward perturbations (Ferber et al. 2002; Tang et al. 1998). It has been shown that strategies to avoid falls with perturbations or when walking on slippery surfaces involve hip and knee movements in order to counteract changes in the center of mass position and speed (Duysens et al. 2008; Redfern et al. 2001;

Address for reprint requests and other correspondence: D. Farina, Dept. of Neurorehabilitation Engineering, Bernstein Focus Neurotechnology Göttingen, Bernstein Center for Computational Neuroscience, University Medical Center Göttingen, Georg-August Univ., Von-Siebold-Str. 4, 37075 Göttingen, Germany (e-mail: dario.farina@bccn.uni-goettingen.de).

You et al. 2001). These relevant mechanisms to avoid falls have been described only for forward perturbations, in the sagittal plane.

Previous evidence suggests participation of afferent information to modulate limb movements (Kargo and Giszter 2000b). Furthermore, limb trajectory might be dependent on the afferent participation for the wiping reflex in frogs, even though the target movement is accomplished, since the combination of spinal motor programs ensures limb motion in deafferented animals (Kargo and Giszter 2000a). In humans, sensory information during walking tunes the muscular activation in order to optimize descending commands (Lacquaniti et al. 2012). Therefore, the modulation of gait in a slippery event might be influenced by the sensory inputs from perturbations, but the original task (gait) might still be maintained since the addition of biomechanical components during walking does not necessarily influence the overall locomotor program (Ivanenko et al. 2005). Therefore, in this study it was hypothesized that the motor modules related to normal locomotion would not be influenced by perturbations, but additional motor modules would be superimposed on the current motor programs in order to assist balance recovery. To test the hypothesis, we analyzed the reaction to recover balance during walking after perturbations in different planes. Slips in the anterior-posterior (AP) and medial-lateral (ML) directions were induced by a moveable force platform embedded in a walkway, and surface electromyography (EMG) was recorded from the lower limb, trunk, and neck muscles.

METHODS

Subjects

Eight healthy men (age 28 ± 4 yr; body mass 71 ± 10 kg; stature 171 ± 7 cm) volunteered for the experiment. They had no known history of neurological or motor disorder at the time of testing that may have affected their ability to perform the experiment. All subjects provided written informed consent before participation, and the procedures were approved by the ethical committee of Northern Jutland (N-20100042).

Walking Conditions

The experiment consisted of repeated walking trials along a 7-m walkway. A force platform was positioned in the middle of the walkway, embedded in the floor. Subjects were asked to step with the right foot on the force platform and to continue walking. The force platform could be activated for translational movements (10-cm translation in 150 ms) at an average velocity of 66.67 cm/s and peak velocity of 88.7 cm/s, in different directions during walking. Lower limb and trunk muscle EMG, joint kinematics, and ground reaction forces were recorded from the full gait cycle preceding heel strike on the platform and from the full gait cycle following heel strike while the subjects walked along the walkway.

Normal (unperturbed) walking. After familiarization, the subjects performed 10 unperturbed trials to establish a normal walking pattern. They were asked to walk at their natural speed along the walkway, at a constant velocity (on average 1.3 ± 0.11 m/s). During these trials they were not informed that the platform could translate but were asked to step on the platform.

Perturbed walking. After the unperturbed gait trials, two sets of trials were performed for the AP (forward/backward) and ML (leftward/rightward) perturbations. The subjects experienced three types of stimuli: 1) catch trials; 2) direction-1 (forward for AP and leftward for ML); and 3) direction-2 (backward for AP and rightward for ML). Forward, leftward, and rightward perturbations were delivered 35 ms after the initial contact, whereas backward perturbations were delayed

by 250–350 ms, as determined by the timing of the stance phase in the prior unperturbed trials. This delay served to deliver the backward perturbations during the late stance phase. During the catch trials, no perturbation was delivered, although the subjects were not informed as to which trials were of this type. For each direction, 12 trials were performed, totaling 30 fully randomized trials for each set (24 perturbed trials and 6 catch trials). A rest interval was provided after each 15 trials in order to avoid fatigue effects. Catch trials were included in the protocol in order to reduce predictability of the perturbations. Figure 1 illustrates the walking tasks performed.

Data Recording

Kinematics. Retroreflective ball-shaped markers were placed bilaterally on each side of the subject on the skin overlying the following landmarks: heel, first and fifth metatarso-phalangeal joint, lateral malleolus, lateral condyle, greater trochanter, anterior superior iliac spine, posterior superior iliac spine, and gleno-humeral joint. Additional markers were also placed in the bilateral segments (foot, shank, thigh, and arms), serving as tracking markers to define the three-dimensional (3D) motion. In addition, markers were placed on the seventh cervical vertebra and upper and bottom ends of the sternum. The marker positions were tracked with a motion analysis system with eight infrared digital video cameras (Oqus 300 series, Qualisys, Gothenburg, Sweden). Kinematic data were recorded with a sampling frequency of 256 Hz and synchronized with the EMG and kinetic recordings. All subjects wore the same type of walking shoes, provided during the experiment, and full-stretch top and pants covering the EMG cables to avoid movement artifacts.

Kinetics. Ground reaction forces were recorded at 1,024 Hz by a 3D force platform (AMTI, OR6-5, Watertown, MA) mounted to a hydraulic actuator (van Doornik and Sinkjaer 2007). Software based on the LabVIEW platform (MrKick II, Aalborg University, Aalborg, Denmark) was used for recording. With an electric feedback circuit, the vertical force (F_z) served as trigger signal to initiate the force plate movement.

Electromyography. Surface EMG signals were recorded in bipolar configuration with Ag/AgCl electrodes (Ambu Neuroline 720 01-K/12; Ambu, Ballerup, Denmark) with 22-mm center-to-center spacing. A reference electrode was placed on the right wrist. Prior to electrode placement the skin was shaved and lightly abraded. The EMG signals were amplified with a gain of 2,000 (EMG-USB, LISiN; OT Bioelettronica, Turin, Italy), band-pass filtered (second-order, zero-lag Butterworth, bandwidth 10–450 Hz), sampled at 2,048 Hz, and A/D converted on 12 bits per sample. The EMG signals were recorded bilaterally from the following muscles according to the SENIAM recommendations (Hermens et al. 2002), except for those explicitly described: peroneus longus (PER), gastrocnemius lateralis (GL), soleus (SOL), tibialis anterior (TA), vastus lateralis (VL), rectus femoris (RF), biceps femoris (BF), tensor fascia latae (TFL), gluteus maximus (GLU), rectus abdominis (RAB, ~3 cm lateral of the umbilicus; Ivanenko et al. 2006), external oblique (EOB, right above the midpoint between the top of the iliac spine and the anterior superior iliac spine; Kaneda et al. 2009), and erector spinae at L1 (ESP, 2 cm lateral to the spinous process; Ivanenko et al. 2006), medial deltoideus (MD), upper trapezius (TRA), splenius capitis (SPL), and sternocleidomastoideus (SCM).

Data Analysis

For the kinematic analysis, the body was modeled as an interconnected chain of rigid segments: foot, shank, thigh, pelvis, and trunk were investigated, with joint angles between segments being analyzed in Visual 3-D (v. 3.79, C-motion). The ipsilateral and contralateral gait cycles under consideration were defined with respect to the right heel strike on the force platform to the subsequent right heel strike. The first heel strike and stance phase were determined from the force plate recordings (when the vertical ground reaction force exceeded 20 N). The platform movement was activated by the trigger signal linked to the force threshold

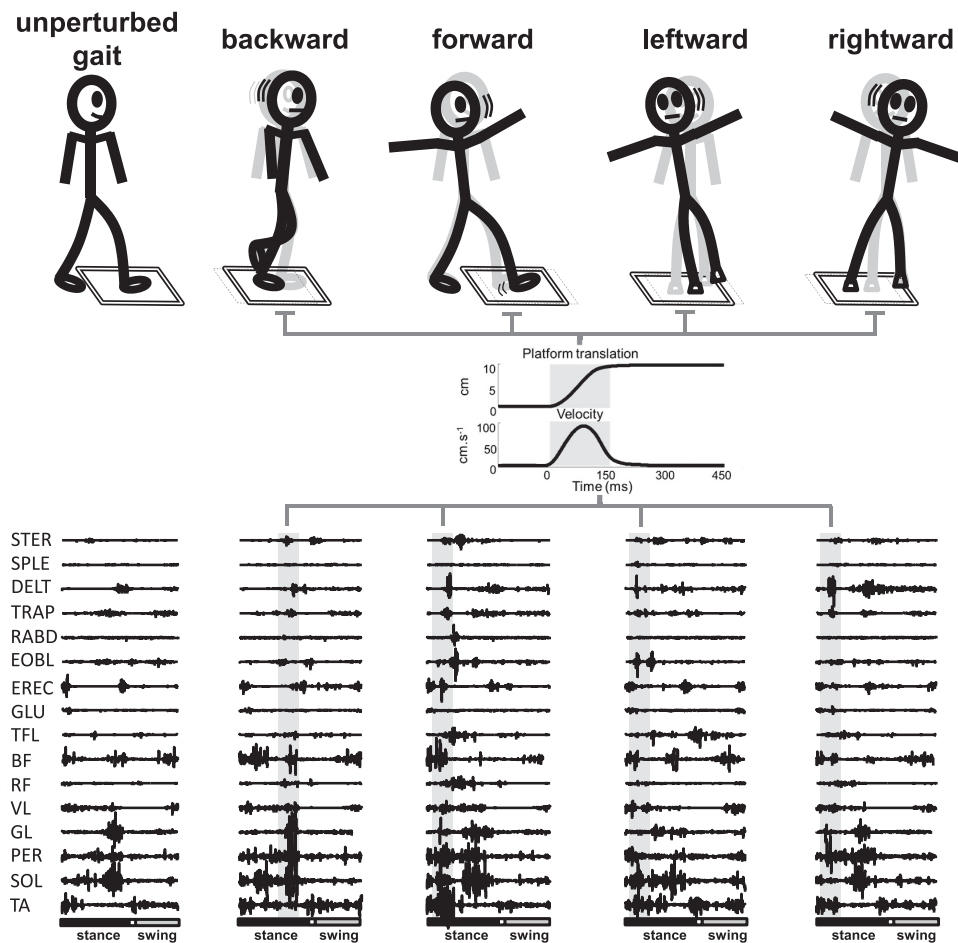


Fig. 1. Representative illustration of walking tasks with raw EMG signals from 1 representative subject. Subjects initially performed unperturbed gait, followed by forward, backward, leftward, and rightward perturbations (in random order). The perturbations were triggered by contact with the ground. Forward, leftward, and rightward perturbations were delivered at the heel contact, while backward perturbations were delayed by 250–350 ms with respect to heel contact in order to coincide with the late stance phase. Perturbations were 10-cm translation and lasted 150 ms.

(>20 N) to deliver perturbations. The subsequent heel strike was defined by assessing the next minimum value for the Z component of the heel marker.

Signal processing. After segmentation, the surface EMG signals from the 32 muscles were full-wave rectified, low-pass filtered (10 Hz), and time-normalized in order to obtain 200 data points for one gait cycle (d'Avella et al. 2003; Ivanenko et al. 2004). For each subject, all trials for a given condition were averaged, followed by the application of nonnegative matrix factorization (NMF) (d'Avella et al. 2003; Gizzi et al. 2011; Lee and Seung 2001; Muceli et al. 2010) in order to identify motor modules and activation signals. Similarly to other multivariate analysis tools (Tresch et al. 2006), NMF is used to factorize the matrix representing the muscular activation pattern into the product of two matrixes that represent the motor modules (synergy matrix) and the activation signals. The factorization is performed by minimizing a cost function under specific updating rules (see Lee and Seung 2001), with the only constraint that the synergy matrix and the activation signals are nonnegative.

Motor module model. The EMG signals recorded from M muscles were indicated as

$$X(k) = [x_1(k), x_2(k), \dots, x_M(k)]^T \quad (1)$$

where $x_m(k)$ is the activity of the m th muscle at time instant k . The activation signals $P(k)$ were indicated as ($N < M$)

$$P(k) = [p_1(k), p_2(k), \dots, p_N(k)]^T \quad (2)$$

The relation between $X(k)$ and $P(k)$ is described as follows:

$$X(k) \approx X_r(k) = S \cdot P(k) \quad (3)$$

where $X_r(k)$ is the muscle activity vector reconstructed by the factorization. In Eq. 3, the EMG $X(k)$ are obtained by linear transformation

of the activation signals $P(k)$ with gain factors s_{mn} . The matrix whose columns are the weights of each activation signal for each muscle is denoted as S in Eq. 3 and will be referred to as the motor module (or synergy) matrix (Lee and Seung 1999).

Dimensionality. The number of motor modules N needed for accurate description of the movement was assessed by the dimensionality analysis proposed by d'Avella et al. (2003). According to this procedure, the quality of reconstruction of the muscle activation pattern is analyzed as a function of the number of modules and the minimum number of modules is identified as the point in which this curve pronouncedly changes its slope (d'Avella et al. 2003). In addition to this criterion, a minimum threshold for reconstruction quality was set at 80%. For quantifying the quality of reconstruction, the estimated muscular activation pattern was compared with the recorded pattern by means of the variation accounted for (VAF) value, defined as the variation that can be explained by the model $VAF = 1 - SSE/SST$, where SSE (sum of squared errors) is the unexplained variation and SST (total sum of squares) is the pooled variation of the data.

Similarities were investigated for motor modules and activation signals for both the perturbed and unperturbed legs. The motor module matrices were compared by computing the scalar product between pairs of columns, normalized by the product of the norms of each column (Muceli et al. 2010). Similarities between activation signals were quantified by the value of the cross-correlation function at zero time lag (Gizzi et al. 2011). The presence of shared modules among different perturbation directions was analyzed by computing the similarities between one module from a given condition and each of the modules from another condition until the best match was found.

After computation of the reconstruction quality, the motor modules for each subject were extracted from the concatenation of all trials in a given

perturbation direction for both the perturbed and unperturbed legs. Similarities among subjects and the presence of shared modules among conditions were assessed from this data set. In addition, the EMG activities from all subjects were concatenated for a given condition, from which motor modules were extracted to represent the whole group of subjects. In this way, all the variability in the data set was taken into account and the presence of shared modules among conditions was analyzed.

Kinematic data were low-pass filtered (6 Hz, second-order, zero-lag Butterworth). The right heel positions in the AP direction were used to determine the stride that contains the step over the platform. Stride duration, stride length, scaled to leg length for each subject, and walking velocity were obtained for each trial. Joint angles for each gait cycle were extracted for ankle, knee, and hip (bilaterally) and trunk and were compared qualitatively with the joint angles from unperturbed gait in order to describe the kinematic changes during perturbations. In addition, the stance duration and double support duration were calculated. To verify the effects of the perturbation direction on the kinematic variables, a one-way ANOVA was used, followed by the Tukey post hoc test when necessary (SPSS 19, SPSS, Chicago, IL). An intertrial analysis was performed in order to verify the effect of accommodation to the task on the EMG signals. For this analysis, the EMG envelopes were normalized by the respective maximum value for each perturbation direction. The EMG envelopes were then averaged over three intervals: the first 50% of the stance phase, from 50% to 100% of the stance phase, and during the full swing phase. The values of averaged EMG envelopes in the three intervals were further averaged over the first three and last three trials for each subject. The resulting EMG activity for the first and last three trials were compared by a paired Student's *t*-test to identify potential differences due to the time at which the trials were performed (beginning or end of the experiment). The significance level was set to $P < 0.05$.

RESULTS

Figure 2 shows the stride duration and stride length for both the perturbed (PRT) and unperturbed (UPRT) limbs. The perturbations had a significant effect on stride duration ($P = 0.025$). Backward perturbations showed shorter stride duration compared

with forward perturbations for PRT and UPRT ($P < 0.05$). Stride length was shorter for leftward and rightward compared with forward perturbations for both limbs ($P < 0.05$). The stride duration for any of the perturbation directions was different from baseline. Walking velocities among tasks were not different from each other (1-way ANOVA, $P > 0.05$; Fig. 2). In particular for PRT, the time of initial double support and perturbed stance phase duration were similar among the perturbations ($P > 0.05$).

Kinematics

Figure 3 shows the joint angles for the perturbed and unperturbed legs, and Fig. 4 shows the trunk kinematics. Changes for a given perturbation direction are associated to the plane of motion, e.g., forward perturbations show changes in the sagittal plane (flexion/extension). Backward perturbations induced pronounced ankle dorsiflexion and trunk flexion during the second part of the stance phase, which were corrected before the swing phase. Forward perturbations elicited immediate knee extension, which was corrected before midstance. Furthermore, hip external rotation and trunk extension were increased during stance. Leftward perturbations provoked immediate changes in the ankle, knee, and hip, especially in the sagittal plane. However, the most pronounced changes in this case were associated to the trunk, which was more extended, bent to the left, and externally rotated during most of the swing phase. Rightward perturbations elicited ankle inversion and hip abduction during the stance phase, as well as trunk extension, mostly during the swing phase.

The unperturbed side was less influenced by perturbations, as expected. There were no relevant changes after the sagittal plane perturbations (forward/backward), whereas perturbations in the frontal plane elicited alterations at the ankle (eversion/inversion, depending on the direction) and hip (adduction/abduction, depending on the direction). The trunk kinematics

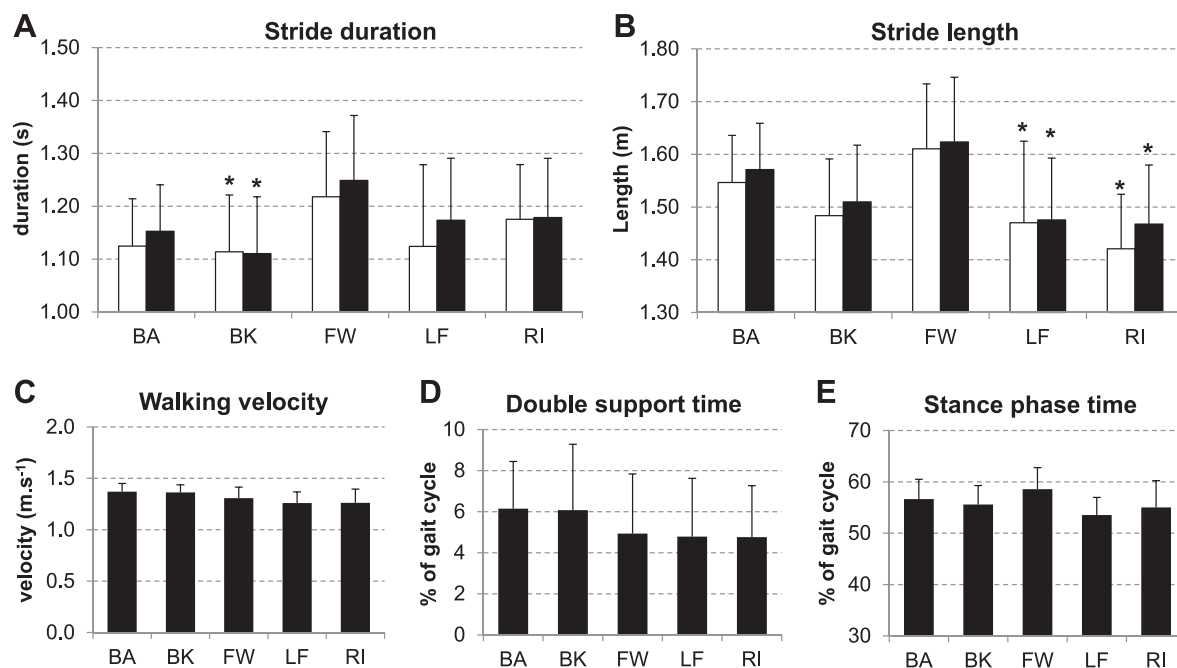


Fig. 2. A and B: mean (SD) stride duration (A) and stride length (B) for both perturbed and unperturbed legs during unperturbed gait (BA) and backward (BK), forward (FW), leftward (LF), and rightward (RI) perturbations. C–E: walking velocity (C), double support duration (D), and stance phase duration (E) for the perturbed leg. *Significant difference with respect to forward perturbation ($P < 0.05$).

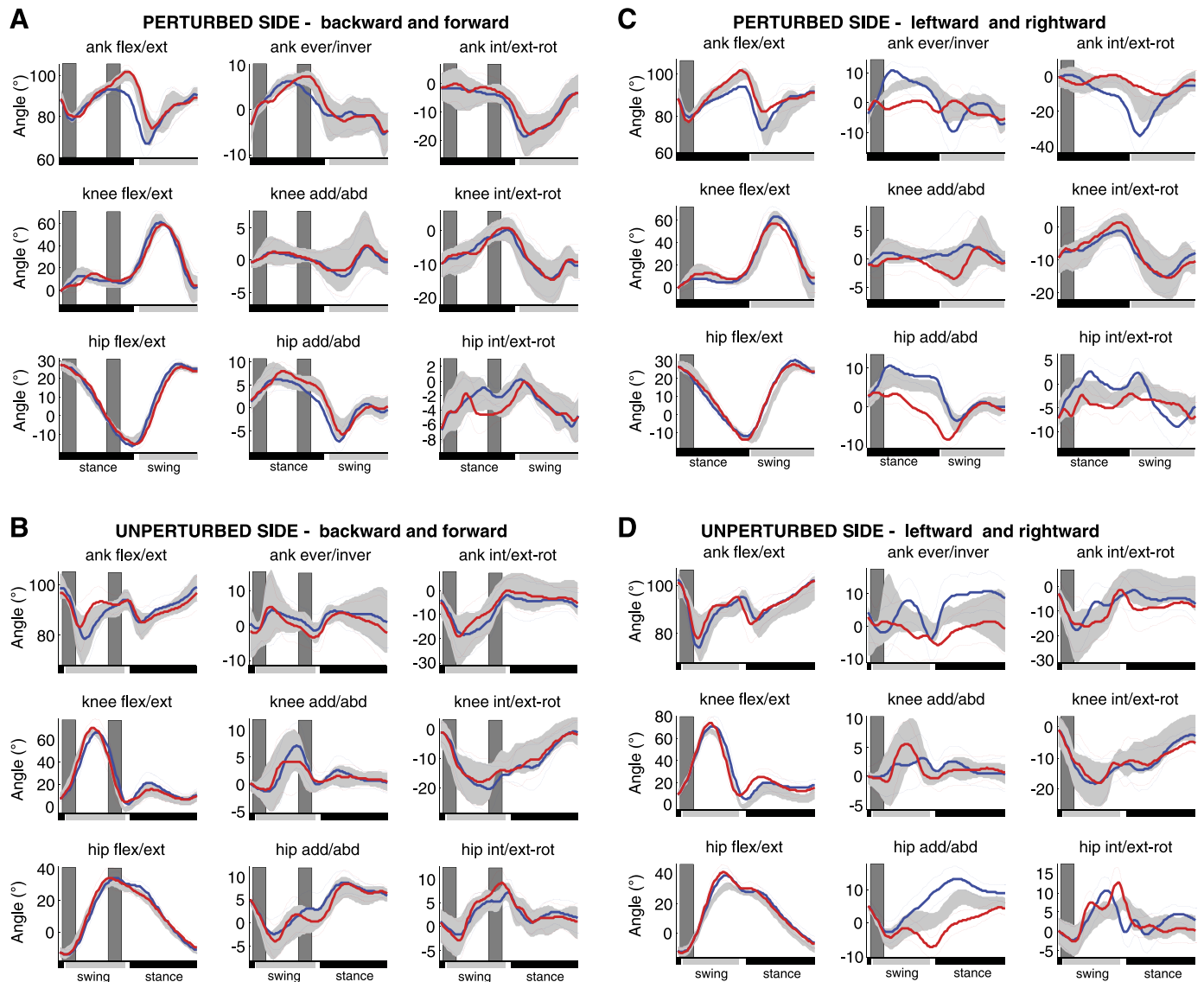


Fig. 3. Averaged perturbed and unperturbed limb kinematics (8 subjects). *Left*: normal walk (light gray area), backward perturbations (blue thick line $\pm$ SD as blue dotted lines), and forward perturbations (red thick line $\pm$ SD as red thin lines) are summarized for perturbed (A) and unperturbed (B) sides. *Right*: normal walk (light gray area), leftward (blue thick line $\pm$ SD as blue dotted lines) and rightward (red thick line $\pm$ SD as red thin lines) perturbations are shown for perturbed (C) and unperturbed (D) sides. Dark gray vertical bars indicate the average perturbation period during the gait cycle. For backward perturbations there is a second dark gray area in A and B, indicating that this perturbation was delivered late during stance phase. Stance and swing phases are indicated as the black and gray areas at *bottom*, and the white region between these areas indicates the amount of variability of the durations.

was also influenced differently depending on the perturbation direction, being usually displaced in the opposite way of the perturbation direction (i.e., trunk moving forward when the perturbation was backward).

Figure 5 shows the center of mass displacement. Forward perturbations elicited changes in the vertical and anterior-posterior component (i.e., lower and forward center of mass) in the early stance phase. Perturbations in the frontal plane (leftward/rightward) induced a change in walking direction. The center of mass was usually in a lower vertical position at the transition from stance to the swing phase (i.e., the minimum value of the curves in Fig. 5A) when sagittal perturbations were compared to unperturbed walking, even though most of the kinematic changes from the sagittal perturbations were counteracted before the swing phase. This means that small changes in the lower limb joints and trunk might still influence the center of

mass positioning, which aids in maintaining safety once the subjects are experiencing perturbations. In the same way, a lower vertical center of mass position at the transition from stance to the swing phase was observed when frontal perturbations were compared to unperturbed walking (Fig. 5B).

Intertrial Variability During Perturbations

Figure 6 shows the comparison between the EMG activity (computed as described in METHODS) from the first three and the last three trials. For most muscles and perturbations, there was no significant effect of the set of trials. There were significant effects only in few cases, especially for the calf muscles, VL and BF, and in these cases the absolute differences were very limited anyway. These results indicate that the learning effect caused by repeated exposure to specific perturbations was only marginal.

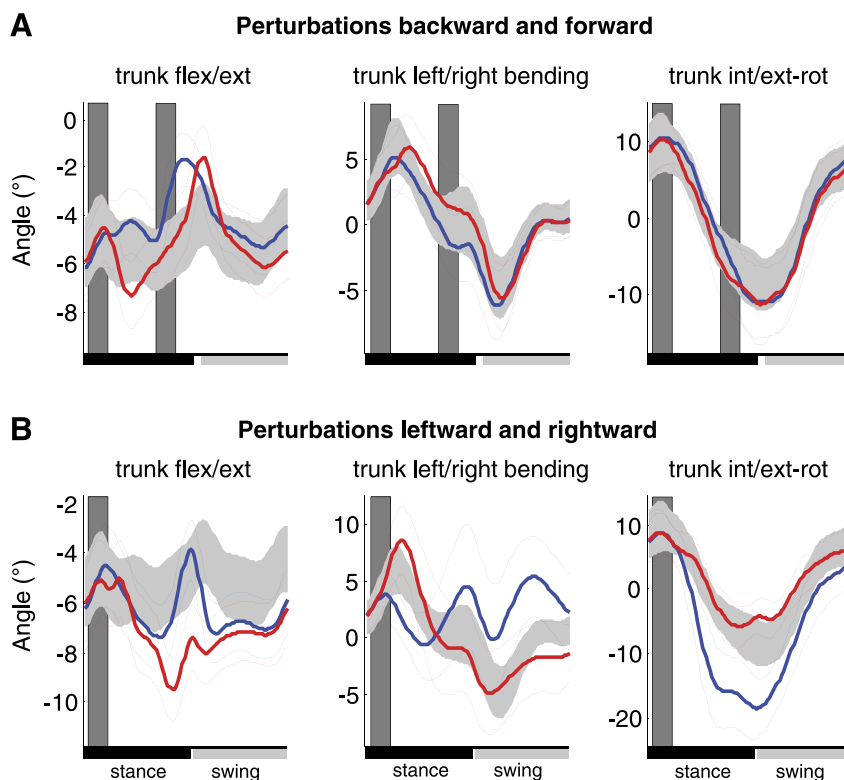


Fig. 4. Averaged perturbed and unperturbed trunk kinematics. **A:** normal walk (light gray area), backward perturbations (blue thick line $\pm$ SD as blue dotted lines), and forward perturbations (red thick line $\pm$ SD as red thin lines). **B:** normal walk (light gray area), leftward perturbations (blue thick line $\pm$ SD as blue dotted lines), and rightward perturbations (red thick line $\pm$ SD as red thin lines). Dark gray vertical bars indicate the average perturbation period during the gait cycle. For backward perturbations there is a second dark gray area in A, indicating that this perturbation was delivered late during stance phase. Stance and swing phases are indicated as the black and gray areas at bottom, and the white region between these areas indicates the amount of variability of the durations.

Number of Motor Modules

The analysis of dimensionality determined that four motor modules were required to reconstruct unilateral muscular activation for both normal and perturbed gait tasks. On average, VAF reached values $>90\%$ (the range of values was $0.82\text{--}0.94\%$) with four modules, and the addition of a fifth module only increased VAF by $5.0 \pm 0.3\%$ (average over all conditions). These results indicate that normal walking and perturbed walking can be expressed by the same number of motor modules.

Description of Perturbed Walking from Motor Modules of Unperturbed Walking

When the four modules obtained from baseline were used to reconstruct the muscular patterns of the perturbed walking, the VAF was on average below 0.8 (VAF backward perturbations = 0.79 ± 0.02 , forward = 0.67 ± 0.1 , leftward = 0.66 ± 0.07 , rightward = 0.75 ± 0.1). This indicated that the modules of normal walking could not explain the perturbed walking. Conversely, the unperturbed leg showed acceptable reconstruction quality with baseline modules, with VAF above 0.8 for all conditions (VAF backward = 0.82 ± 0.1 , forward = 0.82 ± 0.06 , leftward = 0.83 ± 0.14 , rightward = 0.83 ± 0.02).

Intersubject Similarities

The motor modules and activation signals extracted from concatenated trials for different subjects in the different perturbation directions showed a mean similarity among subjects of 0.82 and 0.83, respectively, considering both the perturbed and unperturbed sides. Table 1 shows similarities for motor modules, averaged across the four modules in each perturbation direction. Table 1 also shows similarities between the

activation signals associated to the four modules in each perturbation direction. A high mean similarity (≥ 0.8) was verified among subjects for both the unperturbed and perturbed gait, for both limbs.

Motor Modules

Since the similarity among subjects was high for all conditions for both legs, we extracted motor modules from a concatenation of signals from all subjects in order to take into account all the variability in the data set. Figure 7A shows the muscle weightings for motor modules, and Fig. 8A shows the activation signals for the unperturbed gait and the perturbed gait for each of the perturbation directions (perturbed leg). The four modules found for unperturbed gait (BASE) could be related to the main gait phases. The first module (M1) mainly consisted of the activation of TA and knee extensors and flexor activation during early stance and late swing of normal walking, likely to support the body during load acceptance. The second module (M2) consisted mainly of TFL, EOB, ESP, SPL, and SCM, predominantly active during midstance but also moderately active throughout the swing phase. The third module (M3) consisted mainly of calf muscles (GL and SOL, PER) for body support and forward propulsion. Finally, the fourth module (M4) consisted of TA, RF, and trunk/neck muscles stabilizing the upper body and contributing during the swing phase.

The overall similarity of motor modules from different perturbation directions in comparison to unperturbed gait is shown in Fig. 7A. Walking with backward perturbations was described by similar modules as baseline walking, but the activation signals for M2 (midstance) and especially M4 (swing) were influenced (Fig. 8A), since the perturbation occurred at late stance.

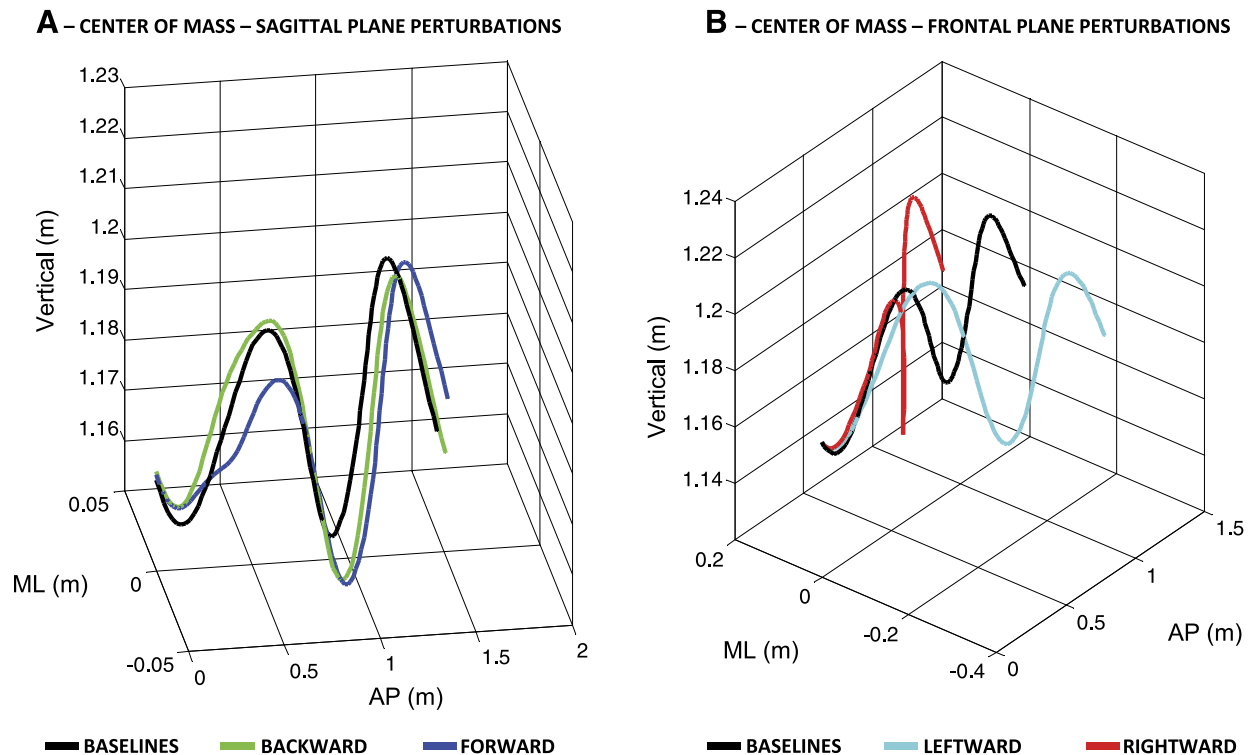


Fig. 5. Three-dimensional illustration of the averaged center of mass displacement for perturbations in the sagittal plane (A) and in the frontal plane (B). All curves start at the same point on *left*. Forward perturbations (blue line in A) show a different pattern in the vertical axis, whereas leftward and rightward perturbations (light blue and red in B) show medial lateral displacements in opposite directions (i.e., perturbations leftward induce walking rightward). ML, medial-lateral; AP, anterior-posterior.

For the other perturbation directions, three of four motor modules were preserved with respect to unperturbed gait (averaged similarity = 0.90 ± 0.07). Depending on the perturbation, one module differed. For forward perturbations, the M1 module (contact) changed, which is explained by a forced knee extension at heel strike, which influences quadriceps activation (VL weighting is reduced) and also elicits reflex components on the ankle muscles in order to increase stiffness. Additionally, this type of perturbation induced a center of mass forward displacement (You et al. 2001), which increased the activation for TA and trunk/neck muscles. In addition to changes in M1, the activation signals for the other motor modules (which were unchanged) were influenced by forward perturbations.

The leftward and rightward perturbations elicited the most substantial changes during swing, where all joints were influenced, together with the trunk. For these perturbations, modules M1, M2, and M3 were preserved but their activation signals changed to fit the new mechanical constraints from a medial or lateral foot displacement. The hip joint was adducted or abducted for leftward and rightward perturbations, respectively, and for both cases there was internal rotation. Similar to the events from forward perturbations, the changes in segment positions during stance required adjustments only to the activation signals. However, the swing phase was compromised by a new walking direction that frontal plane perturbations provoked (see Fig. 5). Thus both module M4 and its activation signal changed in order to move the leg from an unexpected position to perform the swing. Frontal plane perturbations provoked a different swing phase, most likely aiming at preventing a fall instead of maintaining a normal walking pattern.

This could explain the absence of evident activation signals for the swing phase.

The unperturbed leg showed essentially the same modular organization found for normal walking, with no changes in motor modules (average similarity 0.97 ± 0.03) (Fig. 7B). However, the activation signals exhibited differences with respect to unperturbed gait, especially for the initial contact and the swing phase (Fig. 8B).

DISCUSSION

When gait was perturbed in different directions, the modularity of the muscular pattern was maintained and three of four motor modules were common between unperturbed and perturbed gait. These changes were robust across individuals; therefore the neural control of human balance may solve immediate threats during gait by activating specific modules at the most adequate timing. Furthermore, the unperturbed side, which has to perform the crucial next step, showed the same muscle modules for unperturbed and perturbed conditions, even though the planned direction could have changed.

Recent investigations suggested that human locomotion may be controlled by just a few inborn motor modules implemented in a neural network or central pattern generator at the spinal level, which provides the major input to motoneuron activity (Dominici et al. 2011; Ivanenko et al. 2004; Rossignol et al. 2006). The modulation of timing is distributed through the gait cycle in a task-dependent manner, which is determined by the combination of supraspinal and sensory information (Cappellini et al. 2010; Rossignol et al. 2006). Our results are in agreement with these hypotheses, since the perturbed gait

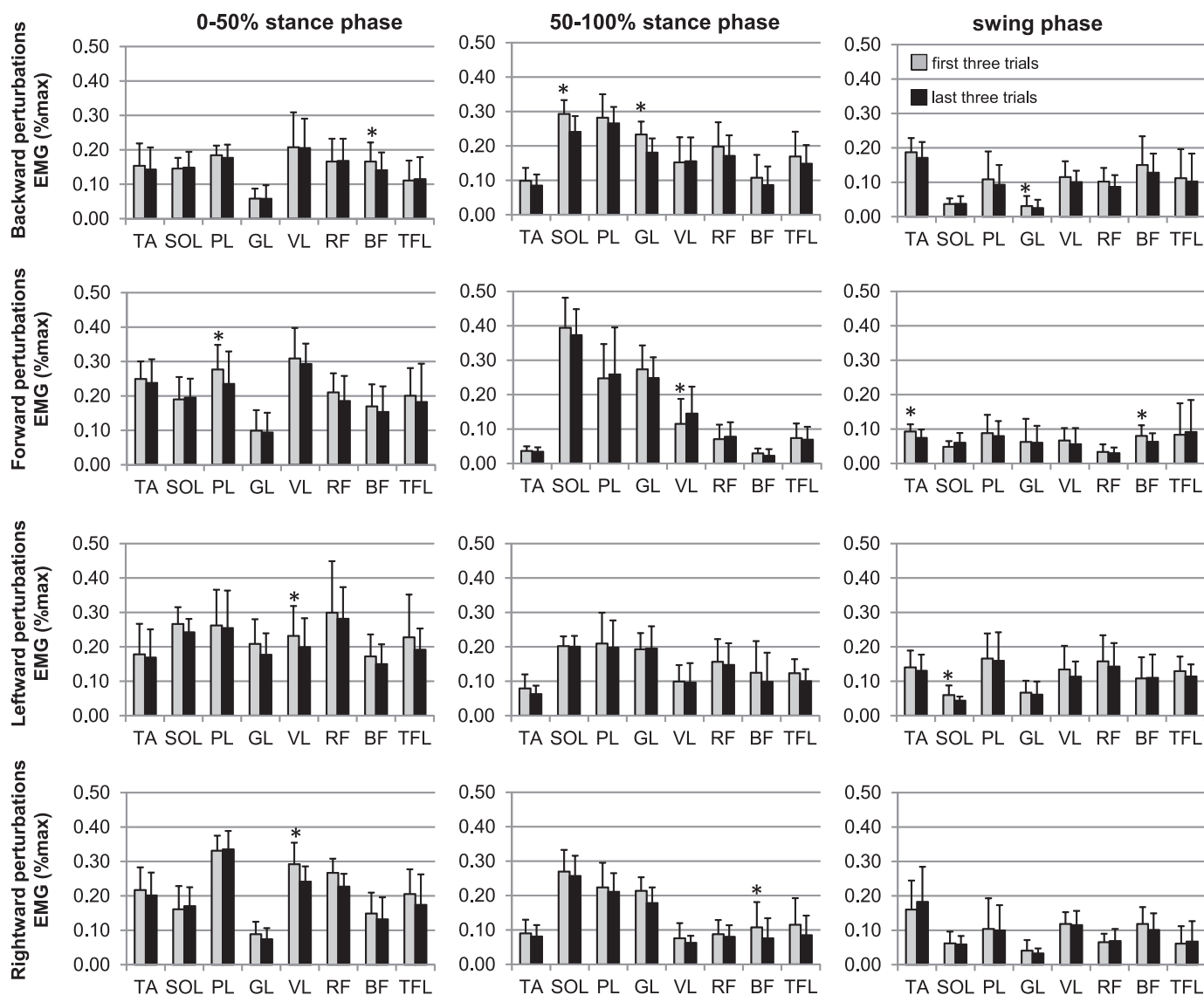


Fig. 6. Mean (SD) EMG activity (see text for details on how this is computed) in 3 time intervals for the 4 perturbation directions [backward (*top*), forward (2nd row), leftward (3rd row), rightward (*bottom*)] for the first 3 perturbed trials (gray bars) and the last 3 perturbed trials (black bars). EMG activity was averaged for the first 50% of the stance phase (*left*), from 50% to 100% of the stance phase (*center*), and during the swing phase (*right*). *Significant difference between the 2 sets of trials ($P < 0.05$).

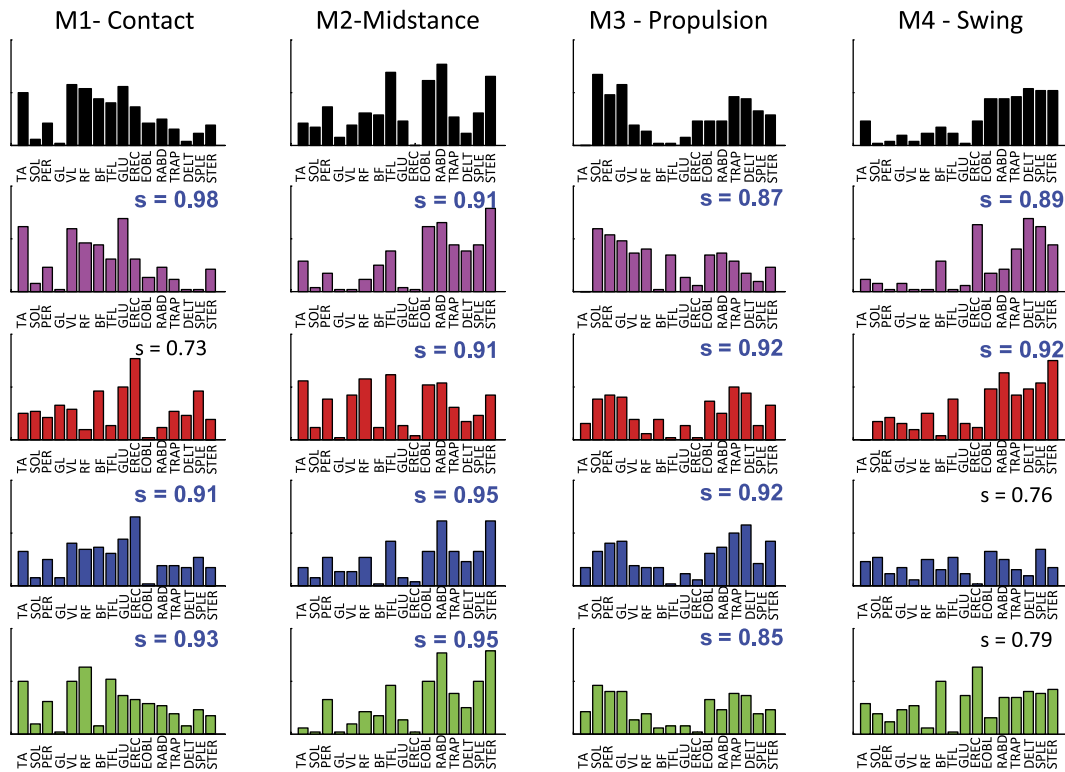
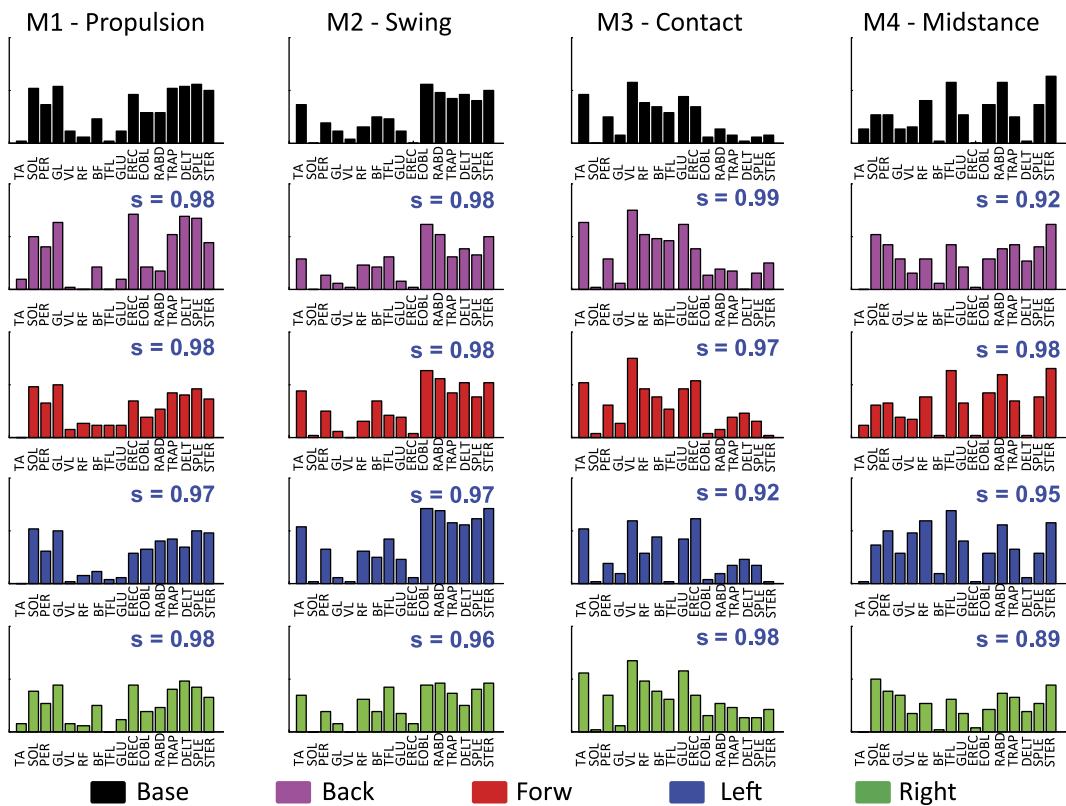
could be explained by most of the motor modules observed during the unperturbed gait, with adjustments in the activation signals. However, postural responses from perturbations may have a more complex interaction, in which brain stem (Macpherson et al. 1997) and cortical (Beloozerova et al. 2003; Mihara et al. 2008) participation are possible. Bhatt and Pai (2009) found that the successful recovery of balance after

experiencing slips requires task-related experience, such that the CNS can continuously modify motor plans. Our results corroborate these findings by indicating that innate locomotion characteristics (motor modules) cannot be drastically changed because of a perturbation, whereas timing patterns are influenced by essential afferent information that updates the slipping lower limb status at the spinal level.

Table 1. Intersubject similarity for motor modules and activation signals of perturbed and unperturbed legs for different perturbation directions and for unperturbed walking

	BASE		BACK		FORW		LEFT		RIGHT	
	PERT	UNP	PERT	UNP	PERT	UNP	PERT	UNP	PERT	UNP
MOD	0.84 ± 0.12	0.80 ± 0.16	0.85 ± 0.1	0.84 ± 0.11	0.84 ± 0.09	0.84 ± 0.21	0.83 ± 0.14	0.80 ± 0.13	0.87 ± 0.09	0.79 ± 0.14
ACT	0.83 ± 0.09	0.76 ± 0.14	0.87 ± 0.08	0.83 ± 0.09	0.83 ± 0.07	0.80 ± 0.1	0.83 ± 0.09	0.80 ± 0.09	0.86 ± 0.05	0.80 ± 0.12

Values are mean ± SD intersubject similarity for the motor modules (MOD) and activation signals (ACT) of the perturbed (PERT) and unperturbed (UNP) legs for different perturbation directions and for unperturbed walking (BASE). BACK, backward; FORW, forward; LEFT, leftward; RIGHT, rightward.

A – MOTOR MODULES – PERTURBED SIDE**B – MOTOR MODULES – UNPERTURBED SIDE**

Base

Back

Forw

Left

Right

Fig. 7. Motor modules for the perturbed (A) and unperturbed (B) sides. *Top to bottom*: baseline, backward, forward, leftward, and rightward conditions. Modules from different conditions are ordered by the events from baseline (*top*, black), except for M4 of leftward perturbations, which has no homologous module in the analysis for the perturbed side. Similarities (s) were inserted for each motor module compared to the corresponding module of baseline. Similarities above 0.80 are highlighted in blue.

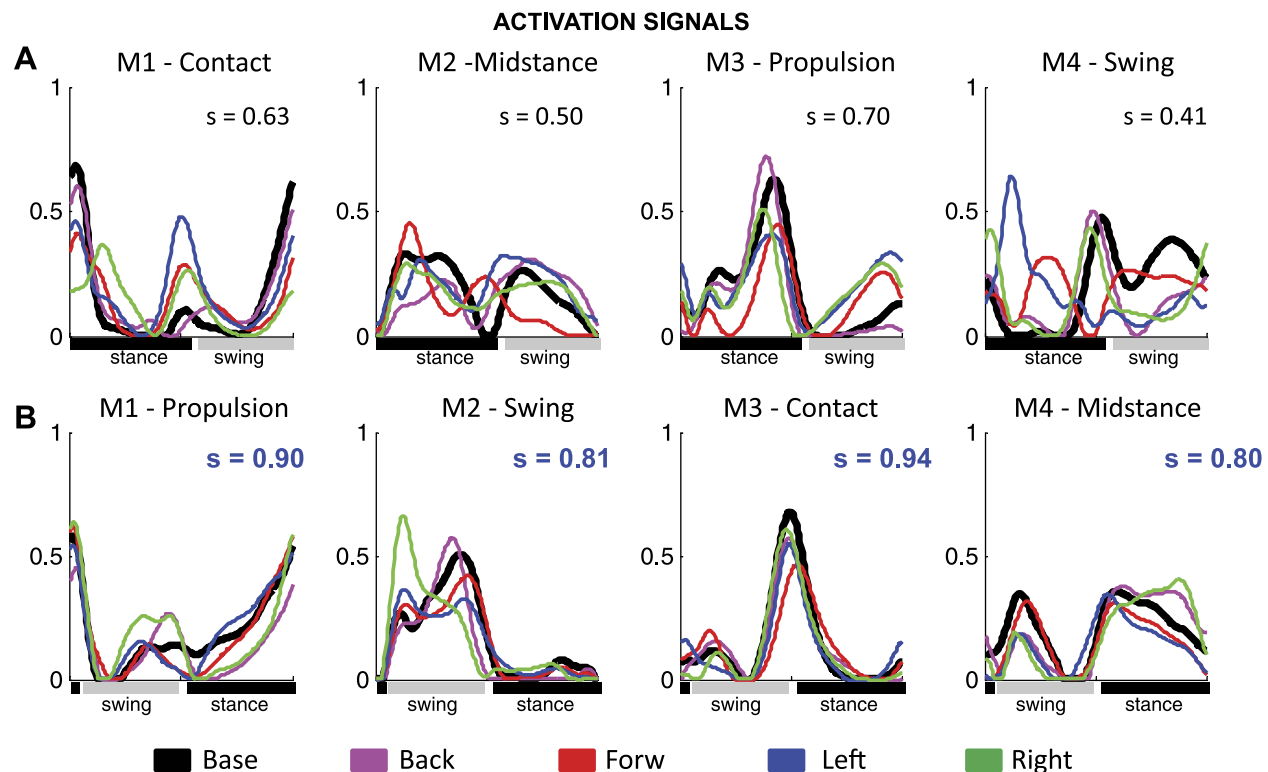


Fig. 8. Activation signals for the perturbed (A) and unperturbed (B) sides associated to the 4 motor modules reported in Fig. 7. Average similarities (s) were inserted for each activation signal compared to the signal of baseline. Similarities above 0.80 are highlighted in blue.

The reduced amount of changes in the unperturbed limb may be due to the reduced afferent information from subcutaneous receptors, muscle spindles, and other structures throughout the perturbed task (Rossignol et al. 2006). There might exist a marginal effect from interlimb neural network pathways, which may elicit contralateral reflex activities (Bhatt and Pai 2009), but our results suggest that changes in the activation signals for the unperturbed leg are under voluntary control rather than an automatic response.

Four modules were sufficient to reconstruct a locomotion task with sufficient quality ($VAF > 0.88$). This result is in agreement with previous studies on cyclic locomotion, which have successfully represented locomotion tasks by using four (Gizzi et al. 2011; McGowan et al. 2010; Monaco et al. 2010;) or five (Ivanenko et al. 2005, 2006) modules. Differences among studies may relate to the specific sets of muscles investigated. Studies conducted by using up to 16 muscles per body side describe gait by four modules, whereas using 32 muscles produced a greater dimensionality (Ivanenko et al. 2004, 2005). In the present study the perturbed walking could also be explained by the same dimensionality as the normal walking. This result suggests that maintenance of balance during perturbed walking may be attained with a similar modular organization as unperturbed walking by preserving most of the motor modules used for unperturbed walking.

Three of the four motor modules used in unperturbed gait were still present when perturbations occurred in different directions. A previous study investigated the control of voluntary tasks during walking (Ivanenko et al. 2005) and verified that additional modules were needed to explain the task. However, our protocol induced changes in locomotion by a perturbation on the foot, rather than a previously planned

motion. The activation of a single different motor module in case of forward, leftward, and rightward perturbations may be compared to the results from multidirectional perturbations while standing investigated by Torres-Oviedo and Ting (2007). The authors verified that balance recovery could be achieved by using a few motor modules (6 or less), which were differently combined to generate postural responses depending on the perturbation direction. In the same way, our results suggest that neural control of human balance during walking is achievable by a low-dimensional set of motor modules that are predominantly similar, regardless of the perturbation direction. The present investigation shows similar conclusions but in a different context in which gait tasks also contain inertial components from the center of mass displacement. Moreover, the center of mass displacement is also combined with a constant modulation for the muscular activation, whereas standing requires equilibrium maintenance until the perturbation occurs.

Postural control during AP perturbations may be described by an “ankle” strategy, which consists of recruiting distal lower limb muscles, whereas a “hip” strategy related to ML perturbations involves more proximal muscles (Horak et al. 1997; Tang et al. 1998; Torres-Oviedo and Ting 2007). Our findings also corroborate previous findings of AP postural responses occurring predominantly more distal. ML perturbations were delivered at heel strike, causing changes especially in the modulation of hip muscles during the stance phase. It remains possible that the overall modulation of the swing is preserved when perturbations do not induce limb dislocation in the sagittal plane, whereas the stance phase becomes compromised by unusual positioning of body segments. Muscles controlled by modules associated with specific biomechanical tasks have

altered mechanical output if the mechanics of the task is changed in simulations (McGowan et al. 2010). For instance, the muscles related to body support modules are sensitive to changes in body weight; therefore it was suggested that motor modules can be modulated independently to produce a coordinated movement to maintain forward progression (McGowan et al. 2010). Moreover, perturbations to balance in different directions might elicit the recruitment of motor modules that allow center of mass acceleration in order to recover balance (Chvatal et al. 2011). The results of the present investigation corroborate these findings, since there were specific changes in the motor modules related to the biomechanical adjustment necessary to keep the forward progression after perturbation, which also may require repositioning of the center of mass. Our results also suggest some flexibility in the muscle weightings, so that the biomechanical task could be achieved (McGowan et al. 2010).

Perturbations forward induce anterior foot displacement, which requires immediate activation of anterior leg muscles and both the anterior and posterior thigh muscles (Tang et al. 1998). Previous studies have shown that the most relevant strategies to regain balance after slipping include knee and hip movements, in order to counteract the center of mass displacement (Cham and Redfern 2001; Duysens et al. 2008). The results of the present study showed that the activation signal related to heel strike is prolonged until the perturbation is finished, which may be an important strategy to maintain balance. The analysis performed in this study does not allow the identification of reflex components, since the filtering used for extracting the EMG envelopes has a relatively low cutoff frequency. Nonetheless, these components are important for the recovery strategy since the increased activation of the anterior leg and thigh muscles in the M2 module occurs at the moment of perturbation, as verified previously when humans are perturbed forward while walking (Duysens et al. 2008).

Leftward perturbations induced more extensive kinematic changes throughout the gait cycle. The activation signals for the M4 module (see Fig. 8A, blue line) suggest that it could be inserted between the contact and midstance modules. As previously explained, it may only modulate global stiffness given the difficulty in finding a solution that provides the transition from the load acceptance to midstance. Results on perturbations in the frontal plane during walking are very scarce (O'Connor and Kuo 2009; Oddsson et al. 2004), with no reference to EMG or 3D kinematics. Therefore it is difficult to compare our findings to previous work. Perturbations in the frontal plane during standing require different strategies to maintain balance compared with the sagittal plane (Jones et al. 2008; Matjacic et al. 2001; Torres-Oviedo and Ting 2007). Our results replenish this information by suggesting similar phenomena for perturbed walking.

A limitation of this study was that subjects were repeatedly exposed to perturbations, which may reduce the initial responses that monosynaptic and polysynaptic reflexes represent to postural corrections (Rossignol et al. 2006). In addition, our elicited perturbations cannot be classified as fully unexpected, since subjects were experiencing these perturbations throughout the experimental protocol. The repeated exposure might also have influenced the activation timing, such as during the first half of the stance phase for backward perturbations, which was expected to be similar to the normal walking since the

perturbation was delayed. This effect may have contributed to the minor changes over time observed in the EMG envelopes (Fig. 6). Nonetheless, the subjects experienced randomized perturbations in different directions interspaced with unperturbed or catch trials without perturbations, and they did not have knowledge on which situation would occur at each trial. In this way, the possibility of anticipation was minimal. The catch trials were previously investigated, and they do not present any changes in kinematics/kinetics and only minor effects on the EMG activity during the stance phase (Oliveira et al. 2012). Another issue might be the fact that subjects were targeting to step onto the platform, even though there was a sufficient familiarization procedure in order to minimize environmental effects. This fact might influence EMG activity, which may not mimic a fully natural gait pattern. However, we found consistent motor modules, which are in line with previous results in the literature from overground walking where no targeting was required.

In conclusion, the muscular activation during walking with perturbations elicited in different directions can be described by a small set of motor modules, which is similar to the set used during normal walking (all modules but one). The strategies to recover balance in the different directions were similar across subjects. Contrary to the initial hypothesis that additional motor modules would be required to assist balance recovery, the results showed that the CNS adapts existing motor modules to achieve the biomechanical goal rather than including new modules. Despite the similarity in motor modules, the activation signals were all substantially influenced by the perturbations, being presumably the result of the integration of afferent information and supraspinal control.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the author(s).

AUTHOR CONTRIBUTIONS

Author contributions: A.S.C.O., L.G., U.G.K., and D.F. conception and design of research; A.S.C.O., L.G., U.G.K., and D.F. performed experiments; A.S.C.O., L.G., U.G.K., and D.F. analyzed data; A.S.C.O., L.G., U.G.K., and D.F. interpreted results of experiments; A.S.C.O., L.G., U.G.K., and D.F. prepared figures; A.S.C.O., L.G., U.G.K., and D.F. drafted manuscript; A.S.C.O., L.G., U.G.K., and D.F. edited and revised manuscript; A.S.C.O., L.G., U.G.K., and D.F. approved final version of manuscript.

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